中山大学本科生期末考试

考试科目:《编译原理》(A卷答案)

学年学期：20182019学年第一学期　　姓　　名：_____

学　院/系：数据科学与计算机学院/软件工程学　　号：_____

考试方式：闭卷　　年级专业：_____

考试时长：120分钟　　班　　别：_____

警 示　《中山大学授予学士学位工作细则》第八条："考试作弊者，不授予学士学位。"

----------以下为试题区域，共九道大题，总分 100 分，考生请在答题纸上作答----------

1. Write regular expressions for the following languages (8 points).

   a) All strings over {0, 1} that do not contain consecutive zeros (4 points).

   (0?1)*0?

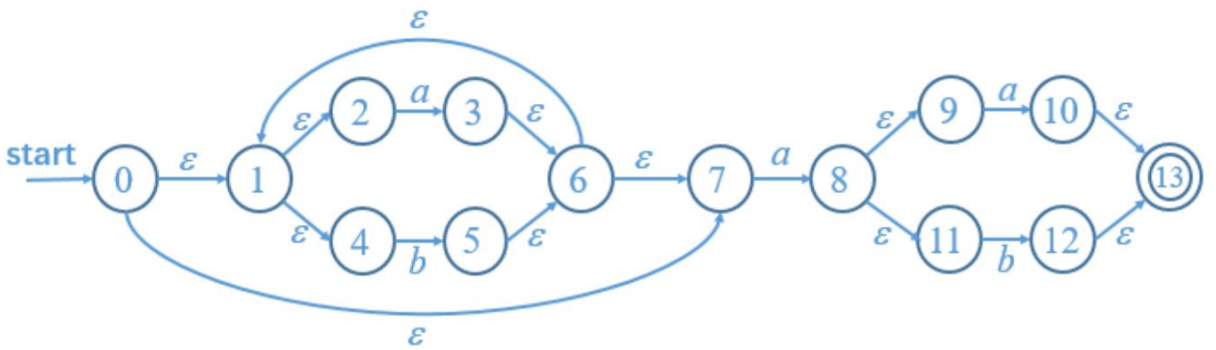
   b) All strings representing a nonnegative binary number (without leading zeros) which is a multiple of 8 (4 points).

   0 | 1(1|0)*000

2. Consider the following regular expression (17 points):

   (a | b) * a (a | b)

   a) Based on the Thompson Algorithm, construct the NFA from the above regular expression (8 points).



b) Convert the NFA into a DFA with minimum number of states (9 points).

-closure({0}) = {0, 1, 2, 4, 7}

-closure(move({0, 1, 2, 4, 7}, 'a')) = -closure({3, 8}) = {1, 2, 3, 4, 6, 7, 8, 9, 11} -closure(move({0, 1, 2, 4, 7}, 'b')) = -closure({5}) = {1, 2, 4, 5, 6, 7}

-closure(move({1, 2, 3, 4, 6, 7, 8, 9, 11}, 'a')) = -closure({3, 8, 10}) = {1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13}

-closure(move({1, 2, 3, 4, 6, 7, 8, 9, 11}, 'b')) = -closure({5, 12}) = {1, 2, 4, 5, 6, 7, 12, 13}

-closure(move({1, 2, 4, 5, 6, 7}, 'a')) = -closure({3, 8}) = {1, 2, 3, 4, 6, 7, 8, 9, 11} -closure(move({1, 2, 4, 5, 6, 7}, 'b')) = -closure({5}) = {1, 2, 4, 5, 6, 7}

-closure(move({1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13}, 'a')) = -closure({3, 8, 10}) = {1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13}

-closure(move({1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13}, 'b')) = -closure({5, 12}) = {1, 2, 4, 5, 6, 7, 12, 13}

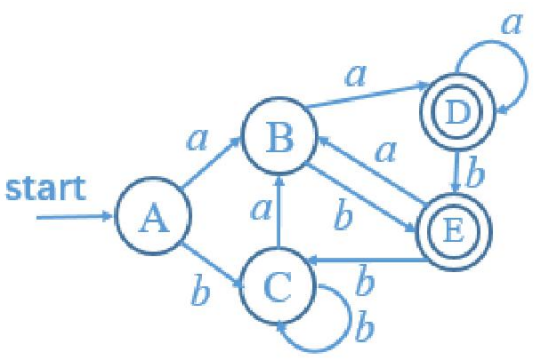
-closure(move({1, 2, 4, 5, 6, 7, 12, 13}, 'a')) = -closure({3, 8}) = {1, 2, 3, 4, 6, 7, 8, 9, 11}

-closure(move({1, 2, 4, 5, 6, 7, 12, 13}, 'b')) = -closure({5}) = {1, 2, 4, 5, 6, 7}

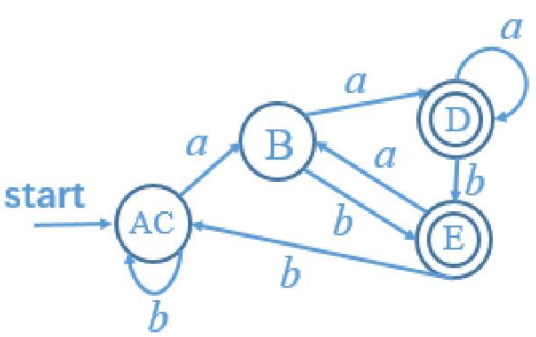
NFA states	DFA state	a	B

{0, 1, 2, 4, 7}	A	B	C

{1, 2, 3, 4, 6, 7, 8, 9, 11}	B	D	E
{1, 2, 4, 5, 6, 7}	C	B	C
{1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 13}	D	D	E
{1, 2, 4, 5, 6, 7, 12, 13}	E	B	C



Minimized DFA:



3. Consider the following CFG (9 points):

$$S \rightarrow +SS \mid -SS \mid a$$

and the string +a-aa.

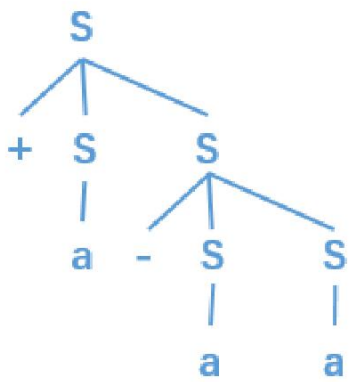
a) Give a leftmost derivation for the string (3 points).

S => +SS => +aS => +a-SS => +a-aS => +a-aa

b) Give a rightmost derivation for the string (3 points).

S => +SS => +S-SS => +S-Sa => +S-aa => +a-aa

c) Give a parse tree for the string (3 points).



4. Write CFGs for the following languages (8 points):

a) L={$a_n b_m c_n$ | m 0, n 1} (4 points).

S → aSc | aAc

A → bA |

b) All strings over {a, b} that begin and end with the same letter (4 points).

S → aAa | bAb | a | b

A → aA | bA |

5. Consider the following CFGs (18 points):

S → i(E)t(E)X | E

X → e(E) |

$E \rightarrow aY \mid bY$

$Y \rightarrow c(E) \mid$

where i, t, e, a, b, c, (and) are terminals.

a) Compute the FIRST and FOLLOW sets for all non-terminals (8 points).

Non-terminal	FIRST	FOLLOW
S	i, a, b	$
X	e,	$
E	a, b	), $
Y	c,	), $

b) Construct an LL(1) parsing table for the grammar (8 points).

	i	t	e	a	b	c	(	)	$
S	S→i(E)t(E)X			S→E	S→E				
X			X→e(E)						X→
E				E→aY	E→bY				
Y						Y→c(E)		Y→	Y→

c) Is this grammar LL(1)? Why? (2 points)

Yes, because no conflicts can be found in the LL(1) parsing table.

6. Consider the following CFG (12 points):

S→aAD | aBe | bBS | bAe

A→g

B→g

D→d | ε

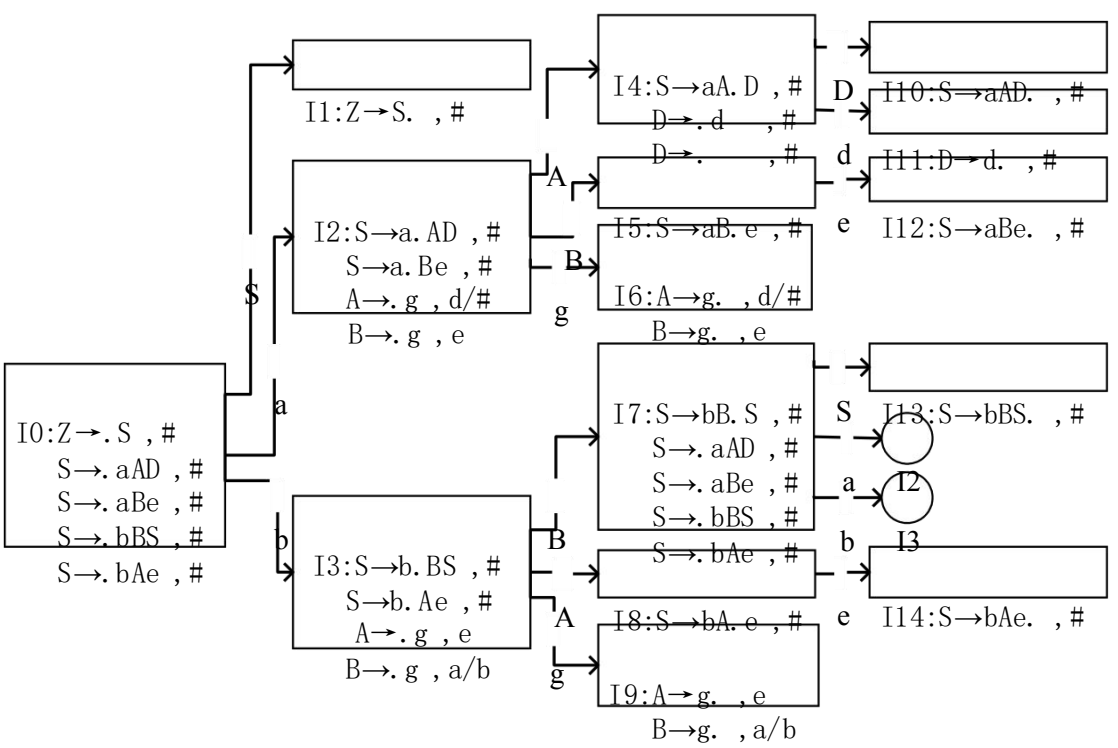
augment the grammar and construct the LR(1) sets of items for the augmented

grammar.

首先拓广 G[S]为 G[Z]：

0：Z → S	1：S → aAD	2：S → aBe
3：S → bBS	4：S → bAe	5：A → g
6：B → g	7：D → d	8：D → ε

构造G[Z]的LR（1）项目族为：



7. The following grammar generates binary strings and their complements (10 points).

F → B
| B

$$
\begin{aligned}
B \rightarrow\ & B0 \\
\mid\ & B1 \\
\mid\ & 0 \\
\mid\ & 1
\end{aligned}
$$

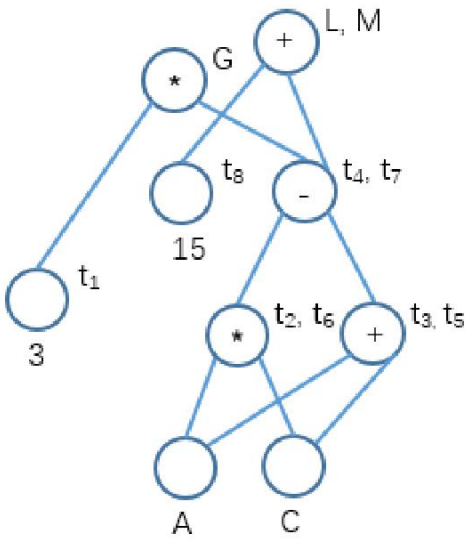
The value of a (non-negated) string is just the decimal value of the binary number the string represents; the value of a negated string is the decimal value of the string with 1's replaced by 0's and 0's replaced by 1's. For example, the value of 010 is 2 and ¬010 is 5. Design a syntax-directed definition (SDD) for the above grammar such that the non-terminal F has an attribute F.val which keeps the value of an input string generated by F. Please do NOT modify the grammar.

产生式	语义规则
$F \rightarrow B$	B.c = false, F.val = B.val
$F \rightarrow \neg B$	B.c = true, F.val = B.val
$B \rightarrow B_1 0$	B_1.c = B.c, B.val = B_1.val * 2 + (B.c ? 1 : 0)
$B \rightarrow B_1 1$	B_1.c = B.c, B.val = B_1.val * 2 + (B.c ? 0 : 1)
$B \rightarrow 0$	B_1.c = B.c, B.val = (B.c ? 1 : 0)
$B \rightarrow 1$	B_1.c = B.c, B.val = (B.c ? 0 : 1)

8. Consider the following basic block (10 points):

1)	$t_1 = 3$
2)	$t_2 = A \quad C$
3)	$t_3 = A+C$
4)	$t_4 = t_2 - t_3$
5)	$G = t_1 \quad t_4$
6)	$t_5 = A+C$
7)	$t_6 = A \quad C$
8)	$t_7 = t_6 - t_5$
9)	$t_8 = t_1 \quad 5$
10	$L = t_8+t_7$
11	$M = L$

a) Construct the DAG of the above basic block (5 points).



b) Assume that only G, L and M will be used after the basic block. Give the

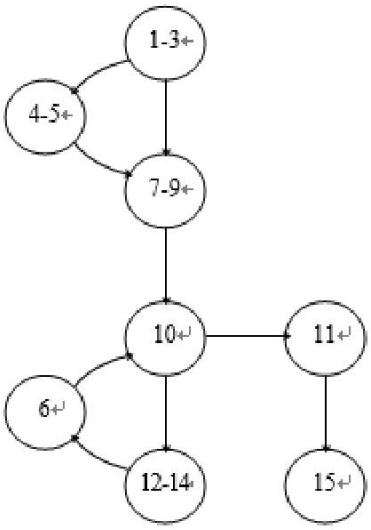
optimized three-address statement sequence (5 points).

1)	$t_2 = A * C$
2)	$t_3 = A + C$
3)	$t_4 = t_2 - t_3$
4)	$G = 3 * t_4$
5)	$L = 15 + t_4$
6)	$M = L$

9. Consider the following fragment of three-address instructions (8 points):

```
(1)          b := 1
(2)          b := 2
(3)          if w <= x goto B
(4)          e := b
(5)          goto B
(6)     A:   goto D
(7)     B:   c := 3
(8)          b := 4
(9)          c := 6
(10)    D:   if y <= z goto E
(11)         goto End
(12)    E:   g := g + 1
(13)         h: = 8
(14)         goto A
(15)    End: h := 9
```

Please partition these three-address instructions into basic blocks, and draw the

control flow graph. You may draw the resulting graph directly, but you must mark

each node by number n~m indicating that the corresponding basic block consists of
instructions n through m, inclusive.